

AhangTech: AI Collaboration & Coding Guardrails

Operational Constraints & Quality Principles for Autonomous & Assisted Development

To maintain system stability, domain integrity, and clean cross-session integration across AI coding agents, the following non-negotiable guardrails are established for the AhangTech codebase:

Guardrail 1: The Contract is Law (Pydantic / OpenAPI)

Pydantic schemas in backend/schemas.py and generated OpenAPI contracts are the single source of truth for data shapes. AI agents must inspect schemas before writing consumers or producers. Never invent fields or alter types quietly. Any schema change requires an explicit issue and documented update in docs/DECISIONS.md.

Guardrail 2: Strict Architectural Decoupling

Backend business logic, ORM models, and database persistence must remain strictly inside backend/. The frontend/ must only communicate with the backend via HTTP REST endpoints. No direct database queries or server dependencies are allowed inside client-side components.

Guardrail 3: Semantic Error Handling & HTTP Statuses

Business state failures must return explicit semantic HTTP codes (e.g., HTTP 400 for checking out an item under repair, 404 for missing entities, 422 for unprocessable payloads). Never return HTTP 200 with embedded error flags.

Guardrail 4: Loud Placeholders Only

AI agents must never silently invent mock data, fake credentials, or assumed construction ratios. Any placeholder must be visibly marked with `PLACEHOLDER_`, `// TODO: verify`, or documented in docs/DECISIONS.md[cite: 1].

Guardrail 5: Zero Unnecessary Bloat

Do not introduce complex state libraries (Redux, Zustand), distributed task runners (Celery), or unrequested authentication layers. Focus purely on fulfilling the current ticket's Definition of Done with standard FastAPI and lightweight UI patterns.

Guardrail 6: Pre-Push Hygiene & Diff Verification

Every AI contribution must pass the project formatter/linter and include at least one unit/smoke test[cite: 1]. Review git diffs carefully before landing changes — never touch files outside the assigned ticket scope[cite: 1].